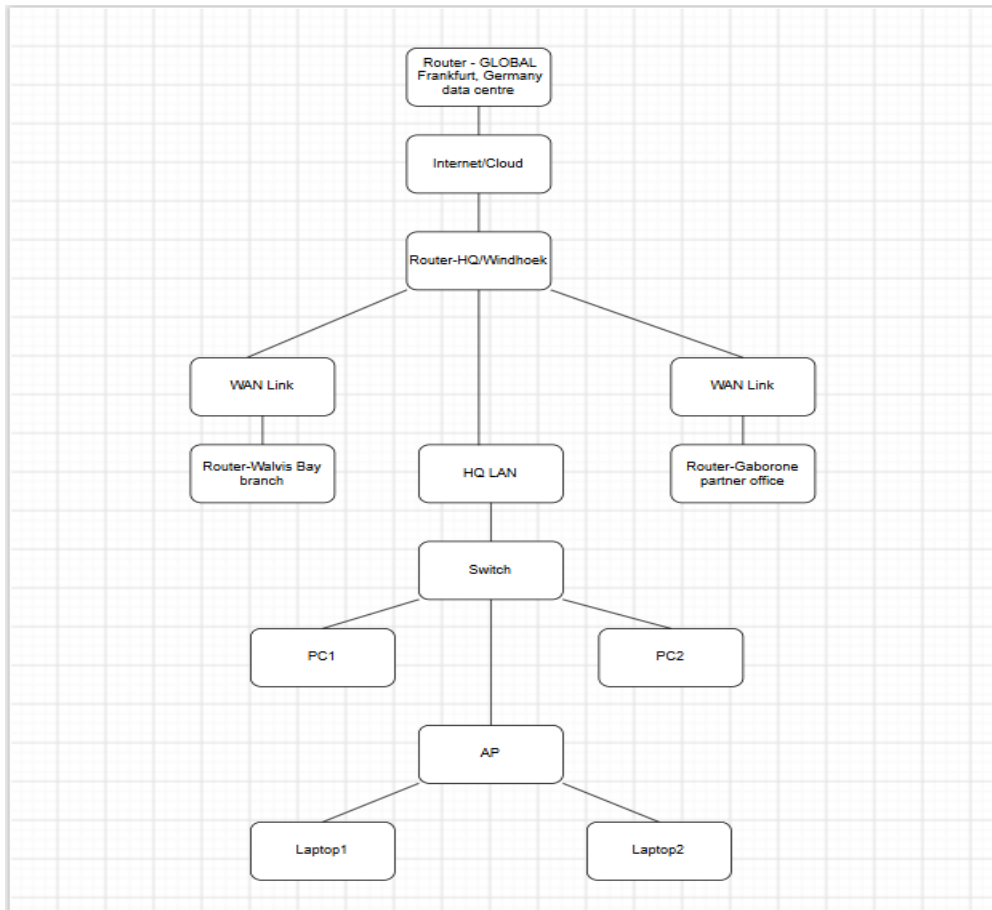


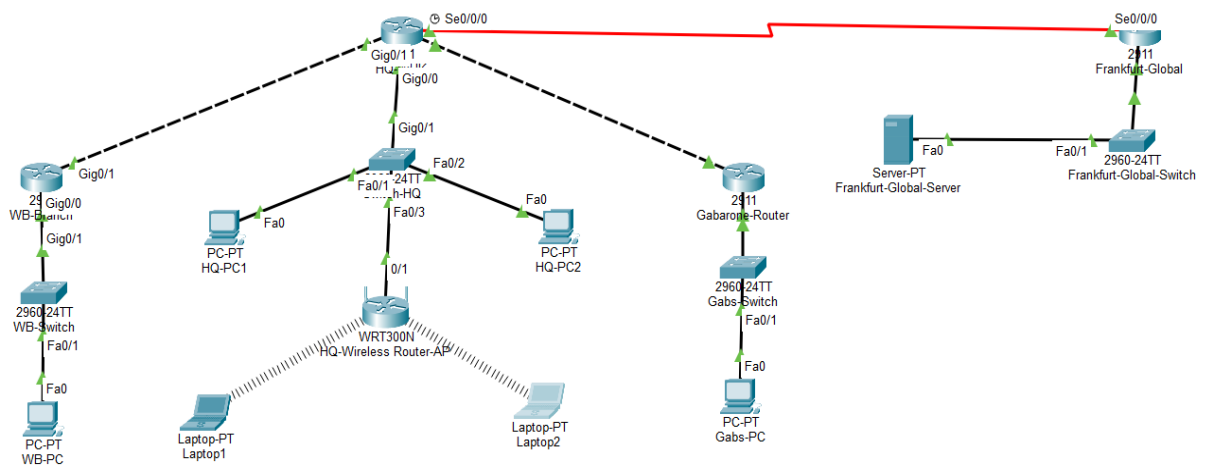
Part B — Practical: Cisco Packet Tracer Simulation

Topology

- **Windhoek**, Namibia is our Headquarters
- **WalvisBay**, Namibia is our Branch
- **Gaborone**, Botswana will be our SADC partner
- **Frankfurt**, Germany is our global data centre.



Topology with all connections including the two wireless laptops connected to the Access point.



Subnet IP addressing to separate the LAN networks at each location. /24 networks were assigned to each site. routers.

Site	Device	IP Address	Subnet Mask	Default Gateway
Windhoek HQ	HQ Router	192.168.10.1	255.255.255.0	NA
Windhoek HQ	PC1	192.168.10.10	255.255.255.0	192.168.10.1
Windhoek HQ	PC2	192.168.10.11	255.255.255.0	192.168.10.1
Windhoek HQ	Laptop 1	192.168.10.20	255.255.255.0	192.168.10.1
Windhoek HQ	Laptop 2	192.168.10.21	255.255.255.0	192.168.10.1
Windhoek HQ	WRT300N	192.168.10.2	255.255.255.0	NA
Walvis Bay	Router	192.168.20.1	255.255.255.0	NA
Walvis Bay	PC	192.168.20.10	255.255.255.0	192.168.20.1
Gaborone	Router	192.168.30.1	255.255.255.0	NA
Gaborone	PC	192.168.30.10	255.255.255.0	192.168.30.1
Frankfurt	Router	192.168.40.1	255.255.255.0	NA
Frankfurt	Server	192.168.40.10	255.255.255.0	192.168.40.1

/30 networks were used for the WAN connections between

WAN Link	Router A	Router B	Subnet
Windhoek / Walvis Bay	10.0.0.1	10.0.0.2	10.0.0.0/30
Windhoek / Gaborone	10.0.0.5	10.0.0.6	10.0.0.4/30
Windhoek / Frankfurt	10.0.0.9	10.0.0.10	10.0.0.8/30

This addressing method keeps each site traffic separate and makes routing so much easier.

Router Configurations

```

HQ-WHK
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

Router>enable
Router#config ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname HQ-WHK
HQ-WHK(config)#interface g0/0
HQ-WHK(config-if)#ip address 192.168.10.1 255.255.255.0
HQ-WHK(config-if)#no shut
HQ-WHK(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
HQ-WHK(config)#interface g0/1
HQ-WHK(config-if)#ip address 10.0.0.1 255.255.255.252
HQ-WHK(config-if)#no shut
HQ-WHK(config-if)#
HQ-WHK(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
HQ-WHK(config)#interface g0/2
HQ-WHK(config-if)#ip address 10.0.0.5 255.255.255.252
HQ-WHK(config-if)#no shut
HQ-WHK(config-if)#
HQ-WHK(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
exit
HQ-WHK(config)#interface serial 0/0/0
HQ-WHK(config-if)#ip address 10.0.0.9 255.255.255.252
HQ-WHK(config-if)#clock rate 64000
HQ-WHK(config-if)#no shut
HQ-WHK(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
HQ-WHK(config-if)#exit
HQ-WHK(config)#interface serial 0/0/0
HQ-WHK(config-if)#ip address 10.0.0.9 255.255.255.252
HQ-WHK(config-if)#clock rate 64000
HQ-WHK(config-if)#no shut
HQ-WHK(config-if)#exit
HQ-WHK(config)#end
HQ-WHK#
%SYS-5-CONFIG_I: Configured from console by console
HQ-WHK#write memory
Building configuration...
[OK]
HQ-WHK#

```

```
Router#
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname WB-Branch
WB-Branch(config)#interface g0/0
WB-Branch(config-if)#ip address 192.168.20.1 255.255.255.0
WB-Branch(config-if)#no shut

WB-Branch(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
WB-Branch(config)#write
^
% Invalid input detected at '^' marker.

WB-Branch(config)#write memory
^
% Invalid input detected at '^' marker.

WB-Branch(config)#end
WB-Branch#
%SYS-5-CONFIG_I: Configured from console by console

WB-Branch#write
Building configuration...
[OK]
```

```
WB-Branch#enable
WB-Branch#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0  192.168.20.1    YES manual up              up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1               unassigned      YES unset  administratively down down
WB-Branch#
WB-Branch#config t
Enter configuration commands, one per line. End with CNTL/Z.
WB-Branch(config)#interface g0/1
WB-Branch(config-if)#ip address 10.0.0.2 255.255.255.252
WB-Branch(config-if)#no shut

WB-Branch(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
WB-Branch(config)#end
WB-Branch#
%SYS-5-CONFIG_I: Configured from console by console

WB-Branch#write
Building configuration...
[OK]
```

 Gabarone-Router

Physical Config CLI Attributes

IOS Command Line Interface

```

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Gabarone-Router
Gabarone-Router(config)#interface g0/0
Gabarone-Router(config-if)#ip address 192.168.30.1 255.255.255.0
Gabarone-Router(config-if)#no shut

Gabarone-Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Gabarone-Router(config)#interface g0/1
Gabarone-Router(config-if)#ip address 10.0.0.6 255.255.255.252
% Ambiguous command: "i address 10.0.0.6 255.255.255.252"
Gabarone-Router(config-if)#no shut

Gabarone-Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Gabarone-Router(config)#end
Gabarone-Router#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]

```

 Gabarone-Router

```

Gabarone-Router>enable
Gabarone-Router#show ip interface brief
Interface                IP-Address      OK? Method Status Protocol
GigabitEthernet0/0       192.168.30.1    YES manual up      up
GigabitEthernet0/1       unassigned      YES unset  up      up
GigabitEthernet0/2       unassigned      YES unset  down    down
Vlan1                    unassigned      YES unset  up      down
Gabarone-Router#
Gabarone-Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Gabarone-Router(config)#interface g0/1
Gabarone-Router(config-if)#ip address 10.0.0.6 255.255.255.252
Gabarone-Router(config-if)#no shut
Gabarone-Router(config-if)#exit
Gabarone-Router(config)#end
Gabarone-Router#
%SYS-5-CONFIG_I: Configured from console by console

Gabarone-Router#write
Building configuration...
[OK]
Gabarone-Router#

```

 Frankfurt-Global

Physical Config CLI Attributes

IOS Command Line Interface

```

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Frankfurt-Global
Frankfurt-Global(config)#interface g0/0
Frankfurt-Global(config-if)#ip address 192.168.40.1 255.255.255.0
Frankfurt-Global(config-if)#no shut

Frankfurt-Global(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Frankfurt-Global(config-if)#exit
Frankfurt-Global(config)#inface serial 0/0/0
^
% Invalid input detected at '^' marker.

Frankfurt-Global(config)#interface serial 0/0/0
Frankfurt-Global(config-if)#ip address 10.0.0.10 255.255.255.252
Frankfurt-Global(config-if)#no shut

Frankfurt-Global(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
exit
Frankfurt-Global(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

Frankfurt-Global(config)#end
Frankfurt-Global#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]

```

PC configurations

HQ-PC1

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0003.E4D7.779D

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.10.10

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::203:E4FF:FED7:779D

HQ-PC2

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0050.0F91.92DE

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.10.11

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::250:FFF:FE91:92DE

WB-PC

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0060.2FE9.B1D5

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.20.10

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::260:2FFF:FEE9:B1D5

Gabs-PC

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ On

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00E0.8F72.A502

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.30.10

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::2E0:8FFF:FE72:A502

Frankfurt-Global-Server

Physical **Config** Services Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

FastEthernet0

Port Status

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ On

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.1686.80A9

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.40.10

Subnet Mask 255.255.255.0

Access point configurations

HQ-Wireless Router-AP

Physical Config **GUI** Attributes

Wireless

Setup

Wireless

Security

Access Restrictions

Applications & Gaming

Basic Wireless Settings

Wireless Security

Wireless Security

Security Mode: WPA2 Personal

Encryption: AES

Passphrase: Kalahari@2026

Key Renewal: 3600 seconds

HQ-Wireless Router-AP

Physical Config **GUI** Attributes

Wireless-N Broadband Router

Setup

Setup

Wireless

Security

Access Restrictions

Applications & Gaming

Basic Setup

DDNS

MAC Address Clone

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers)

Host Name:

Domain Name:

MTU: Size: 1500

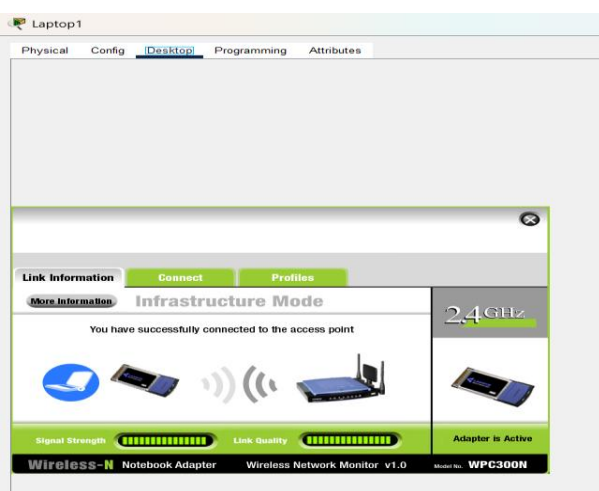
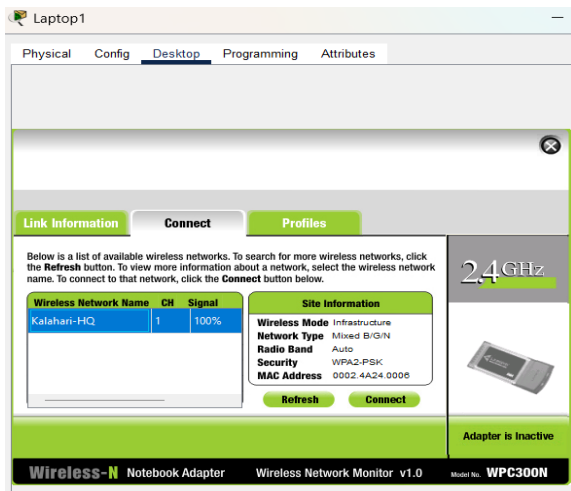
Network Setup

Router IP

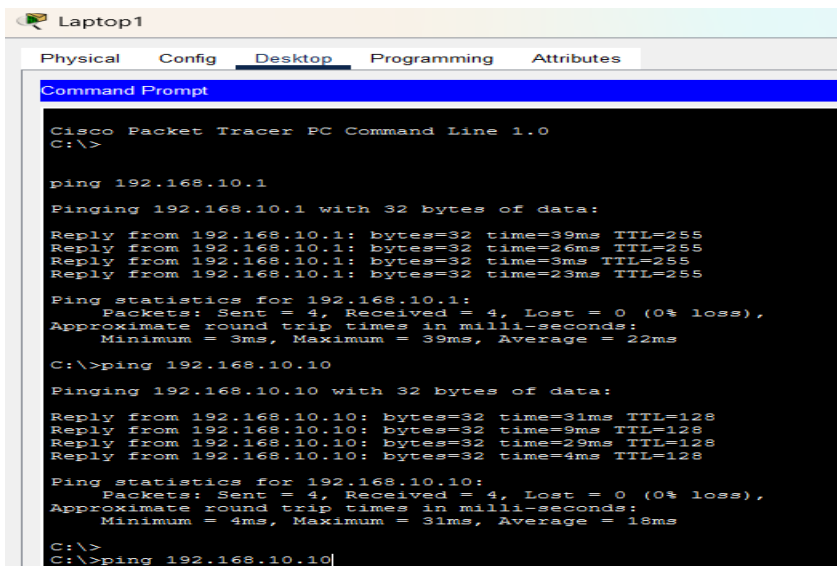
IP Address: 192 . 168 . 10 . 2

Subnet Mask: 255.255.255.0

Successful wireless connectivity



HQ LAN and wifi working



Routing configurations on Windhoek HQ router.

```
HQ-WHK
HQ-WHK>enable
HQ-WHK#config t
Enter configuration commands, one per line. End with CNTL/Z.
HQ-WHK(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
HQ-WHK(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.6
HQ-WHK(config)#ip route 192.168.40.0 255.255.255.0 10.0.0.10
HQ-WHK(config)#end
HQ-WHK#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
HQ-WHK#
```

Routing. Walvis branch now knows that traffic for the other three networks should go through Windhoek HQ.

```
WB-Branch
WB-Branch>enable
WB-Branch#config t
Enter configuration commands, one per line. End with CNTL/Z.
WB-Branch(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
WB-Branch(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.1
WB-Branch(config)#ip route 192.168.40.0 255.255.255.0 10.0.0.1
WB-Branch(config)#end
WB-Branch#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
WB-Branch#
```

Copy Paste

☐ Top

```
Gabarone-Router
write
Building configuration...
[OK]
Gabarone-Router#enable
Gabarone-Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Gabarone-Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.5
Gabarone-Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.5
Gabarone-Router(config)#ip route 192.168.40.0 255.255.255.0 10.0.0.5
Gabarone-Router(config)#end
Gabarone-Router#
%SYS-5-CONFIG_I: Configured from console by console
end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
```


Frankfurt-Global

```
% Invalid input detected at '^' marker.

Frankfurt-Global>enable
Frankfurt-Global#config t
Enter configuration commands, one per line. End with CNTL/Z.
Frankfurt-Global(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.9
Frankfurt-Global(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.9
Frankfurt-Global(config)#ip route 192.168.30.0 255.255.255.0 10.0.0.9
Frankfurt-Global(config)#end
Frankfurt-Global#
%SYS-5-CONFIG_I: Configured from console by console

Frankfurt-Global#write
Building configuration...
[OK]
Frankfurt-Global#
```

Communication works over all networks

HQ-PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=1ms TTL=126
Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time=1ms TTL=126
Reply from 192.168.40.10: bytes=32 time=2ms TTL=126
Reply from 192.168.40.10: bytes=32 time=2ms TTL=126
Reply from 192.168.40.10: bytes=32 time=7ms TTL=126

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 3ms
```

Wireless HQ laptop-to-WAN test

The image displays four screenshots from Cisco Packet Tracer, arranged in a 2x2 grid. The top row shows the configuration for Laptop1 and Laptop2, and the bottom row shows the command prompt outputs for the same devices.

Laptop1 Configuration:

- Physical Tab:** Shows the laptop icon.
- Config Tab:** Shows the configuration for the Wireless0 interface.
 - Port Status:** 300 Mbps
 - MAC Address:** 0003 E43A 8CA0
 - SSID:** Kalahari-HQ
 - Authentication:** WPA2-PSK (selected), WEP, WPA, 802.1X, WPA2, Method.
 - Encryption Type:** AES
 - IP Configuration:** Static (selected), DHCP.
 - IPv4 Address:** 192.168.10.20
 - Subnet Mask:** 255.255.255.0
 - IPv6 Configuration:** Automatic (selected).

Laptop2 Configuration:

- Physical Tab:** Shows the laptop icon.
- Config Tab:** Shows the configuration for the Wireless0 interface.
 - Port Status:** 300 Mbps
 - MAC Address:** 00E0 8F6B 1785
 - SSID:** Kalahari-HQ
 - Authentication:** WPA2-PSK (selected), WEP, WPA, 802.1X, WPA2, Method.
 - Encryption Type:** AES
 - IP Configuration:** Static (selected), DHCP.
 - IPv4 Address:** 192.168.10.21
 - Subnet Mask:** 255.255.255.0
 - IPv6 Configuration:** Automatic (selected).

Laptop1 Command Prompt:

```
C:\>ping 192.168.30.10 with 32 bytes of data:
Reply from 192.168.30.10: bytes=32 time=10ms TTL=126
Reply from 192.168.30.10: bytes=32 time=14ms TTL=126
Reply from 192.168.30.10: bytes=32 time=7ms TTL=126
Reply from 192.168.30.10: bytes=32 time=24ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 24ms, Average = 13ms

C:\>ping 192.168.40.10
Reply from 192.168.40.10: bytes=32 time=20ms TTL=126
Reply from 192.168.40.10: bytes=32 time=10ms TTL=126
Reply from 192.168.40.10: bytes=32 time=24ms TTL=126
Reply from 192.168.40.10: bytes=32 time=9ms TTL=126

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 24ms, Average = 15ms

C:\>tracert 192.168.30.10
Tracing route to 192.168.30.10 over a maximum of 30 hops:
  0  18 ms  5 ms  5 ms  192.168.10.1
  1  7 ms  4 ms  10 ms  10.0.0.6
  2  22 ms  17 ms  20 ms  192.168.30.10
Trace complete.
C:\>
```

Laptop2 Command Prompt:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>

tracert 192.168.30.10

Tracing route to 192.168.30.10 over a maximum of 30 hops:

  1  30 ms  14 ms  23 ms  192.168.10.1
  2   8 ms   7 ms  10 ms  10.0.0.6
  3   8 ms  20 ms   9 ms  192.168.30.10

Trace complete.
C:\>
```

Network Design Justification

For this project, we designed a network for Kalahari Logistics that connects the main office in Windhoek to three other locations. These are the branch in Walvis Bay, the partner office in Gaborone, Botswana, and the global office in Frankfurt, Germany.

At the Windhoek office, we used a Cisco switch to connect the wired computers. We also added a WRT300N wireless router so that two laptops could connect to the network using Wi-Fi. We called the wireless network Kalahari-HQ and added a password to protect it. We used static IP addresses for the devices because it makes it easier to know which IP address belongs to each device and makes troubleshooting easier.

For the connection between Windhoek and Walvis Bay, we used a WAN connection to represent a leased line between the two offices. This would be a good option for a company that needs a reliable connection between its offices in Namibia.

The Gaborone connection represents the company's connection to its regional partner in Botswana. We gave this connection its own IP network so that the Gaborone network and the Windhoek network remain separate but can still communicate with each other.

For the Frankfurt connection, we used a serial WAN connection in Packet Tracer to represent a long-distance connection such as VSAT, satellite or another international WAN service. This was suitable for the project because Frankfurt is outside Africa and represents the company's global connection.

We configured static routes on the routers so that they know where to send traffic when communicating with the other locations. Each site also has its own IP address range, which helps keep the network organised.

After completing the configuration, we tested the network by using ping commands between the different locations. The tests were successful, including communication between Windhoek, Walvis Bay, Gaborone and Frankfurt. We also tested the two wireless laptops and confirmed that they could connect to the Kalahari-HQ Wi-Fi network.

Overall, the Packet Tracer simulation shows that the four locations can communicate with each other and that both wired and wireless connections are working as planned.